PEDRO MORALES

PERSONAL

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EDUCATION

Purdue University August 2022 - Present
Ph.D. in Mathematics
Mentor: Dr. Kiril Datchev

University of Maryland, College Park August 2019 - May 2021
Bachelor of Science, Mathematics

Montgomery College September 2017 - May 2019
Associate of Science, Mathematics

RESEARCH INTERESTS

Partial Differential Equations, Scattering Theory, Wave Propagation, Asymptotic Analysis.

SELECTED PUBLICATIONS

Publications

1. **Aguillon Y, Burkholder E, Cheng X, Eddins S, Harrell E, Kozai K, Leake E, Morales P.** *Linking number of monotonic cycles in random book embeddings of complete graphs* J. Knot Theory Ramif. (2023)
2. **Aguillon Y, Cheng X, Eddins S, Morales P.** *The Mean Sum of Squared Linking Numbers of Random Piecewise-Linear Embeddings of K_n* Rose-Hulman Undergraduate Mathematics Journal. (2023)

Preprints

1. **Christiansen T, Datchev K, Morales P, Yang M.** *Unconditional wave decay in dimension two* arXiv:2507.03140v1. (2025) Submitted for publication.

TEACHING

1. Spring 2026 MA162 *Plane Analytic Geometry And Calculus II* (Recitation Instructor)
2. Fall 2025: MA527 *Advanced Mathematics For Engineers And Physicists I* (Grader)
3. Fall 2024, Spring 2024, Spring 2025: MA261 *Multivariate Calculus* (Recitation Instructor)
4. Fall 2022: MA162 *Plane Analytic Geometry And Calculus II* (Recitation Instructor)

PRESENTATIONS

1. Ohio River Analysis Meeting 2026, University of Kentucky March 2026
2. Graduate Student Analysis Seminar, Purdue University January 2025

3. MAA MathFest 2021 - Session 3 - Applied Mathematics (Online) August 2021.
4. 2020 REU Conference, Indiana University, Bloomington (Online), July 2020
5. Joint Mathematics Meeting 2020, Denver, CO. January 2020

AWARDS

Purdue Alliance Scholar Fellowship August 2022 - August 2024
Highly competitive and prestigious award given to outstanding applicants to Purdue University graduate programs.

CONFERENCES ATTENDED

- Ohio River Analysis Meeting 2026 - ORAM 15, University of Kentucky. March 21-22, 2026
- PDE Mini-School: "On Hilbert's sixth problem", UNC Chapel Hill. February 12-13, 2026
- Field of Dreams Conference, Atlanta, GA November 8-10, 2024
- Triangle Area Inverse Problems Weekend, North Carolina State University. October 4-6, 2024
- Field of Dreams Conference, Georgia State University. November 3-5, 2023
- MAA MathFest 2021 (Online). August 2021
- 2020 REU Conference at Indiana University, Bloomington (Online). July 2020
- Joint Mathematics Meeting 2020, Denver, CO. January 2020

ACADEMIC SERVICE

Purdue University

- Co-organizer of the **Graduate Student Analysis Seminar**, Purdue University 2025.
- Represented the Mathematics department at the **Field of Dreams conference** 2023 and 2024 during the Graduate School Fair.
- Mentored three prospective graduate students through the **mentorship program** of the **Association of Women in Mathematics** (Purdue Chapter).